The Policy Gradient Theorem

Infinite-Horizon Expected Discounted Reward

Sai Sampath Kedari
sampath@umich.edu

Contents

1 The Objective and Why Its Gradient Is Hard **2**
1.1 What θ Controls . . . 2
1.2 Following a Single Step, and Then the Whole Trajectory . . . 2
1.3 Notation and Dimensions . . . 3

2 Differentiating the Value: Two Terms **3**

3 Unrolling the Recursion **5**

4 From the Unrolled Gradient to the Visitation Form **5**
4.1 The Row Vector $\rho_0^\top P_{\pi_\theta}^t$ . . . 6
4.2 Carrying the Sum Through, Entry by Entry . . . 6
4.3 Normalizing the Weights into a Distribution . . . 7

5 The Theorem **7**
5.1 From a Sum to an Expectation: the Likelihood-Ratio Step . . . 7

6 Reading the Gradient **8**

7 Where the Visitation Shift Went **9**
7.1 The Term We Were Afraid Of . . . 9
7.2 What the Derivation Actually Produced . . . 10
7.3 Channel 2 Did Not Vanish: It Became the Upgrade from r to Q . . . 10
7.4 Why This Is Legitimate, and What It Buys . . . 10

A The Matrix Expansion: $\rho_0^\top P_{\pi_\theta}^t$ as a Row of State Probabilities **12**

B The Discounted Visitation Measure $\eta(s)$ **13**

1 The Objective and Why Its Gradient Is Hard

We have a policy $\pi_\theta(a \mid s)$ parameterized by $\theta \in \mathbb{R}^{d'}$, a fixed start-state distribution ρ_0, and the discounted objective

$$J(\theta) \;=\; \sum_{s_0} \rho_0(s_0)\, V^{\pi_\theta}(s_0).$$

The objective is a single number, $J(\theta) \in \mathbb{R}$, so its gradient $\nabla_\theta J(\theta) \in \mathbb{R}^{d'}$ is a vector with one partial derivative per parameter. We want $\nabla_\theta J(\theta)$ so we can move θ in the direction that raises the return. Before deriving anything, let us slowly see what one small change to θ does. That is what makes the gradient look hard.

1.1 What θ Controls

The parameter θ controls one thing: the policy. For each state s, the policy $\pi_\theta(\cdot \mid s)$ is a probability distribution over the actions. Since it is a distribution, the probabilities add up to one. So if we change θ and push more probability onto one action, that probability has to come from somewhere: the other actions lose exactly as much as this one gains. The total stays one; the mass just moves around between actions.

That is the only thing θ does. In particular, θ does not appear in the environment. The environment is the rule $p(s', r \mid s, a)$: given a state s and an action a, it tells us the chance of each next state s' and reward r. That rule belongs to the world, not to us. We cannot change it, and there is no θ inside it. The only knob we have is θ, and it is connected only to the policy.

1.2 Following a Single Step, and Then the Whole Trajectory

Sit at a state s and watch what happens in one step.

- We pick an action from our policy: $a \sim \pi_\theta(\cdot \mid s)$. This is the only place θ is used.
- We give that action to the environment, and the environment gives us back a reward r and the next state s'.

Now change θ a little. The probabilities over actions at s move, so we are likely to pick a different action than before. We give this different action to the environment, and the environment gives us back a different reward and a different next state. The environment did nothing new. It followed the same rule as always. We just gave it a different action, and a different action leads to a different result. So changing *which action we pick* is what changed *where we land.*

This does not stop after one step. The next state s' is a new place where the policy picks an action, and the policy uses the same θ we just changed. So at s' we again pick a slightly different action, and again land in a slightly different state s'', and the same thing happens at s'', all the way down the chain. One small change to θ therefore changes the whole path: every state we visit later, and every reward we collect along the way, shifts. And $V^{\pi_\theta}(s_0)$, the value we are trying to differentiate, is just the discounted sum of all those rewards over the whole path. This is why θ reaches into the entire future.

The two effects, and why the gradient looks hard. Changing θ has two effects. The first is *which actions we pick*: this is $\nabla_\theta \pi_\theta(a \mid s) \in \mathbb{R}^{d'}$, the gradient of the policy, which we can compute exactly because we built the policy. The second is *which states we end up in*: by changing our actions we change where the chain goes, so we change how often we visit each state over the whole future. That second effect is the worry, because "where we go" came out of the environment $p(s', r \mid s, a)$, which we do not know. It looks like the gradient must contain a piece measuring how our visits shift, and measuring that seems to need the environment, which we cannot differentiate. If that were really required, policy gradient would be impossible without a model of the world.

The policy gradient theorem says this worry is unfounded. We will show that $\nabla_\theta J$ comes out using only $\nabla_\theta \pi_\theta$, the easy effect, weighted by how often we visit each state, with no derivative of the environment and no derivative of the visits anywhere in it. The change in where we go is real, but we never have to differentiate it; we get it for free by running the policy and seeing where we end up. The rest of this report makes that precise.

1.3 Notation and Dimensions

$$V^{\pi_\theta}(s) = \sum_a \pi_\theta(a \mid s)\, Q^{\pi_\theta}(s,a), \qquad Q^{\pi_\theta}(s,a) = r(s,a) + \gamma \sum_{s'} p(s' \mid s,a)\, V^{\pi_\theta}(s').$$

With $\gamma < 1$ and an infinite horizon, $Q^{\pi_\theta}(s,a)$ is time-independent: from any state the remaining horizon is always infinite, so "steps left" never changes with the clock, and we may carry a single $Q^{\pi_\theta}(s,a)$ with no time index throughout.

It is worth fixing the dimensions once, because the parameter $\theta \in \mathbb{R}^{d'}$ makes every gradient a d'-vector. The policy value, action-value, and reward, that is $\pi_\theta(a \mid s)$, $V^{\pi_\theta}(s)$, $Q^{\pi_\theta}(s,a)$, and $r(s,a)$, are all scalars in $\mathbb{R}$. Differentiating any of them with respect to θ produces a d'-vector:

$$\nabla_\theta \pi_\theta(a \mid s),\ \nabla_\theta V^{\pi_\theta}(s),\ \nabla_\theta Q^{\pi_\theta}(s,a) \ \in\ \mathbb{R}^{d'}.$$

The objective $J(\theta) \in \mathbb{R}$ is a scalar, so $\nabla_\theta J(\theta) \in \mathbb{R}^{d'}$. We will never fuss over row-versus-column orientation: $\nabla_\theta V^{\pi_\theta}(s)$, $\phi_\theta(s)$, and $\nabla_\theta J$ are all elements of $\mathbb{R}^{d'}$, and when we stack per-state quantities into matrices the *state* index becomes the rows.

Episodic tasks fit this framework without change: model an absorbing terminal state as a self-loop with zero reward. The infinite-horizon discounted formulation then still applies, and $Q^{\pi_\theta}(s,a)$ remains time-independent for stationary policies, so everything below holds for episodic problems as a special case.

2 Differentiating the Value: Two Terms

Differentiate $V^{\pi_\theta}(s) = \sum_a \pi_\theta(a \mid s)\, Q^{\pi_\theta}(s,a)$ by the product rule. Both factors carry θ, so (each term a d'-vector)

$$\nabla_\theta V^{\pi_\theta}(s) = \underbrace{\sum_a \nabla_\theta \pi_\theta(a \mid s)\, Q^{\pi_\theta}(s,a)}_{\textbf{(A)} \in \mathbb{R}^{d'}} + \underbrace{\sum_a \pi_\theta(a \mid s)\, \nabla_\theta Q^{\pi_\theta}(s,a)}_{\textbf{(B)} \in \mathbb{R}^{d'}}.$$

In (A) the scalar $Q^{\pi_\theta}(s,a)$ weights the d'-vector $\nabla_\theta \pi_\theta(a \mid s)$; in (B) the scalar $\pi_\theta(a \mid s)$ weights the d'-vector $\nabla_\theta Q^{\pi_\theta}(s,a)$.

What (A) and (B) mean. (A) freezes every downstream action-value $Q^{\pi_\theta}(s,a)$ as a fixed number and asks: as θ moves and probability mass slides between actions, how much does the average value at s change from that re-aiming of mass alone? This is the *reinforce* term: push mass toward high-value actions. **(B)** freezes the action probabilities at s and asks: how much does the value of the future itself move, because Q^{π_θ} is computed under the same θ we are perturbing? This is the downstream ripple, the same "where we go" worry from Section 1, now appearing inside the derivative.

Now expand (B). In $Q^{\pi_\theta}(s,a) = r(s,a) + \gamma \sum_{s'} p(s' \mid s,a) V^{\pi_\theta}(s')$ the reward r and the transition p are environment quantities with no θ in them, so they differentiate to zero, leaving a d'-vector:

$$\nabla_\theta Q^{\pi_\theta}(s,a) = \gamma \sum_{s'} p(s' \mid s,a)\, \nabla_\theta V^{\pi_\theta}(s') \ \in \ \mathbb{R}^{d'}.$$

Term (B) contains **no derivative of the environment**. Differentiating Q^{π_θ} produces another $\nabla_\theta V^{\pi_\theta}$ at the next states, not a $\nabla_\theta p$. The dynamics appear only as an averaging weight, never differentiated, because θ enters Q^{π_θ} only through the future policy. The chain rule therefore lands on $\nabla_\theta V^{\pi_\theta}(s')$, the same object one step later, and never flows through the transition. This is the first sign that the theorem will not require differentiating the transition model. (It still depends on the environment through Q^{π_θ} and the visitation; what it avoids is $\nabla_\theta p$. In model-free algorithms Q^{π_θ} and the visitation are estimated from sampled trajectories rather than computed from $p(s' \mid s,a)$.)

Substitute (B) back and define the policy-induced one-step transition and the *local reinforce vector*

$$P_{\pi_\theta}(s' \mid s) := \sum_a \pi_\theta(a \mid s)\, p(s' \mid s,a) \ \in \ \mathbb{R}, \qquad \phi_\theta(s) := \sum_a \nabla_\theta \pi_\theta(a \mid s)\, Q^{\pi_\theta}(s,a) \ \in \ \mathbb{R}^{d'}.$$

$\phi_\theta(s)$ is exactly term (A) at state s: the gain in value from re-aiming the action probabilities at s, with every downstream Q^{π_θ} held fixed. It is the local reinforce signal, one d'-vector per state, but "local" here means local in *where the derivative lands*: the derivative hits only $\pi_\theta(a \mid s)$ at the current state. It is *not* local in reward information, because the weight $Q^{\pi_\theta}(s,a)$ already carries the entire discounted downstream future. So $\phi_\theta(s)$ differentiates only the present action choice, yet scores each action by its full future value.

The differentiation now collapses to a recursion in the unknown function $\nabla_\theta V^{\pi_\theta}$, one d'-vector equation per state:

$$\nabla_\theta V^{\pi_\theta}(s) = \phi_\theta(s) + \gamma \sum_{s'} P_{\pi_\theta}(s' \mid s)\, \nabla_\theta V^{\pi_\theta}(s') \qquad (\in \mathbb{R}^{d'}).$$

It is cleanest to read this in stacked form. For each state $\nabla_\theta V^{\pi_\theta}(s)$ and $\phi_\theta(s)$ are vectors in $\mathbb{R}^{d'}$; stacking them over the $|\mathcal{S}|$ states gives matrices

$$\nabla_\theta V \in \mathbb{R}^{|\mathcal{S}| \times d'}, \qquad \phi_\theta \in \mathbb{R}^{|\mathcal{S}| \times d'}, \qquad P_{\pi_\theta} \in \mathbb{R}^{|\mathcal{S}| \times |\mathcal{S}|},$$

where row s of $\nabla_\theta V$ is $\nabla_\theta V^{\pi_\theta}(s)^\top$, row s of ϕ_θ is $\phi_\theta(s)^\top$, and P_{π_θ} has entry (s,s') equal to $P_{\pi_\theta}(s' \mid s)$. Each row of P_{π_θ} is a probability distribution over next states, so P_{π_θ} is *row-stochastic* (every row is nonnegative and sums to one). Crucially, the transition matrix multiplies on the *state* index only; the parameter index d' rides along untouched. The recursion is then

$$\nabla_\theta V = \phi_\theta + \gamma\, P_{\pi_\theta}\, \nabla_\theta V, \qquad \nabla_\theta V,\ \phi_\theta \in \mathbb{R}^{|\mathcal{S}|\times d'}, \quad P_{\pi_\theta} \in \mathbb{R}^{|\mathcal{S}|\times|\mathcal{S}|}.$$

The shapes check: $P_{\pi_\theta}\, \nabla_\theta V$ is $(|\mathcal{S}| \times |\mathcal{S}|)(|\mathcal{S}| \times d') = |\mathcal{S}| \times d'$, matching ϕ_θ. This is a fixed-point equation for the matrix $\nabla_\theta V$: the same unknown appears on both sides, exactly like a Bellman equation but for the gradient of the value instead of the value itself.

3 Unrolling the Recursion

The recursion $\nabla_\theta V = \phi_\theta + \gamma\, P_{\pi_\theta}\, \nabla_\theta V$ is linear in $\nabla_\theta V$. Substitute it into its own right-hand side once, then again, and after K rounds

$$\nabla_\theta V = \sum_{t=0}^{K} \gamma^t\, P_{\pi_\theta}^t\, \phi_\theta \; + \; \gamma^{K+1} P_{\pi_\theta}^{K+1} \nabla_\theta V,$$

where every term is an $|\mathcal{S}| \times d'$ matrix. The last term is the leftover tail. Bound it in the matrix max-norm $\|A\|_\infty = \max_i \sum_j |A_{ij}|$ (the largest absolute row sum), which is submultiplicative:

$$\left\|\gamma^{K+1} P_{\pi_\theta}^{K+1} \nabla_\theta V\right\|_\infty \; \le \; \gamma^{K+1} \left\|P_{\pi_\theta}^{K+1}\right\|_\infty \left\|\nabla_\theta V\right\|_\infty \; = \; \gamma^{K+1} \left\|\nabla_\theta V\right\|_\infty \; \xrightarrow[K\to\infty]{} \; 0,$$

where $\|P_{\pi_\theta}^{K+1}\|_\infty = 1$ because P_{π_θ} is row-stochastic (a product of row-stochastic matrices is row-stochastic), and $\gamma^{K+1} \to 0$ because $\gamma < 1$. Equivalently, this is just the scalar-case bound applied to each of the d' parameter columns of $\nabla_\theta V$ separately. (The standing assumptions that make this clean: finite state and action spaces, bounded rewards, $\gamma < 1$, a policy smooth in θ, and the resulting $\nabla_\theta V$ bounded. This is the only place infinite-horizon discounting is used in the unrolling: it is what kills the leftover.) What remains is the geometric solution

$$\nabla_\theta V = \sum_{t=0}^{\infty} \gamma^t\, P_{\pi_\theta}^t\, \phi_\theta \qquad (\in \mathbb{R}^{|\mathcal{S}|\times d'}).$$

(B) is just (A) at the next state. The ripple introduced no new kind of object. $\nabla_\theta V^{\pi_\theta}(s')$ inside (B) opens, by the same product rule, into "(A) at s'" plus "(B) at s'," and (B) at s' reaches one step further. Unrolling forever turns the entire downstream ripple into a discounted sum of the local reinforce term ϕ_θ, evaluated at every state and weighted by the probability of reaching it. The whole gradient has been brought back to (A)-terms.

4 From the Unrolled Gradient to the Visitation Form

Project the objective onto the start distribution. The start distribution is a row vector $\rho_0^\top \in \mathbb{R}^{1\times|\mathcal{S}|}$ (nonnegative, entries summing to one), and differentiating $J(\theta) = \sum_{s_0} \rho_0(s_0) V^{\pi_\theta}(s_0)$ gives $\nabla_\theta J = \rho_0^\top \nabla_\theta V$, a product of shapes $(1 \times |\mathcal{S}|)(|\mathcal{S}| \times d') = 1 \times d'$, which we read as the gradient vector $\nabla_\theta J \in \mathbb{R}^{d'}$. Substituting the unrolled solution $\nabla_\theta V = \sum_{t\ge 0} \gamma^t P_{\pi_\theta}^t \phi_\theta$ and carrying the row vector inside the sum:

$$\nabla_\theta J(\theta) = \rho_0^\top\, \nabla_\theta V = \rho_0^\top \Big[\sum_{t=0}^{\infty} \gamma^t P_{\pi_\theta}^t\Big] \phi_\theta = \Big[\sum_{t=0}^{\infty} \gamma^t\, \rho_0^\top P_{\pi_\theta}^t\Big] \phi_\theta.$$

Everything now hinges on the row vector $\rho_0^\top P_{\pi_\theta}^t$, the start distribution pushed t steps forward through the chain.

4.1 The Row Vector $\rho_0^\top P_{\pi_\theta}^t$

$P_{\pi_\theta}^t$ is the t-step transition matrix: its (i, j) entry is $\mathbb{P}(S_t{=}s_j \mid S_0{=}s_i)$. Left-multiplying by the start distribution marginalizes out the starting state and leaves the unconditional time-t probabilities (the full bookkeeping is in Appendix A):

$$\rho_0^\top P_{\pi_\theta}^t = \big[\,\mathbb{P}(S_t{=}s_1) \;\; \mathbb{P}(S_t{=}s_2) \;\; \cdots \;\; \mathbb{P}(S_t{=}s_n)\,\big] \;\in\; \mathbb{R}^{1\times|\mathcal{S}|},$$

a row vector whose j-th entry is the probability of sitting in s_j at time t, started from ρ_0 and run under π_θ. (Here $n = |\mathcal{S}|$.) Note this row vector is purely about the *state* index; the parameter dimension d' does not enter until we hit ϕ_θ.

4.2 Carrying the Sum Through, Entry by Entry

Now substitute this row vector back and follow the discounted sum all the way into the final product. The discount γ^t is a scalar, so it multiplies each entry of the row vector, and the sum over t acts entrywise:

$$\nabla_\theta J(\theta) = \Big[\sum_{t=0}^{\infty} \gamma^t \,\big[\mathbb{P}(S_t{=}s_1) \;\; \cdots \;\; \mathbb{P}(S_t{=}s_n)\big]\Big]\,\phi_\theta$$

$$= \underbrace{\Big[\; \underbrace{\textstyle\sum_{t=0}^{\infty} \gamma^t \mathbb{P}(S_t{=}s_1)}_{\eta(s_1)} \;\; \cdots \;\; \underbrace{\textstyle\sum_{t=0}^{\infty} \gamma^t \mathbb{P}(S_t{=}s_n)}_{\eta(s_n)} \;\Big]}_{1\times|\mathcal{S}|} \underbrace{\begin{bmatrix} \phi_\theta(s_1)^\top \\ \phi_\theta(s_2)^\top \\ \vdots \\ \phi_\theta(s_n)^\top \end{bmatrix}}_{|\mathcal{S}|\times d'} = \sum_s \eta(s)\,\phi_\theta(s) \;\in\; \mathbb{R}^{d'}.$$

The left bracket is a row of $|\mathcal{S}|$ scalar weights; the right block is the matrix ϕ_θ, whose row s is the d'-vector $\phi_\theta(s)^\top$. Their product is $1 \times d'$, the gradient $\nabla_\theta J$. Define the scalar weight

$$\eta(s) := \sum_{t=0}^{\infty} \gamma^t \,\mathbb{P}(S_t{=}s) \;\in\; \mathbb{R},$$

so the result is the η-weighted sum of the per-state reinforce vectors, $\nabla_\theta J(\theta) = \sum_s \eta(s)\,\phi_\theta(s)$.

What the product is saying. The row vector $\big[\eta(s_1) \cdots \eta(s_n)\big]$ holds, for each state, the discounted total time the chain spends there (a scalar). The matrix below it holds, in row s, the local reinforce vector $\phi_\theta(s) \in \mathbb{R}^{d'}$. The product scales each state's reinforce vector by how much (discounted) time we spend in that state and sums the results into the single gradient vector $\nabla_\theta J \in \mathbb{R}^{d'}$.

Why the parameter dimension just rides along. Fix one parameter coordinate θ_k and look only at the k-th entry of every $\phi_\theta(s)$. You get back a single scalar dot product $\sum_s \eta(s)\,\phi_{\theta,k}(s)$, which is exactly the k-th component of $\nabla_\theta J$. The vector statement is just these d' scalar dot products run in parallel. The visitation weights $\eta(s)$ are *shared* across all d' of them, because the chain, and therefore η, does not depend on which coordinate of θ we differentiate. This is why the whole derivation could be read as the scalar-θ case repeated d' times: the state index is where all the work happens; the parameter index never couples through the dynamics.

4.3 Normalizing the Weights into a Distribution

The weights $\big[\eta(s_1)\ \cdots\ \eta(s_n)\big]$ are nonnegative but do not sum to one, so they form a *measure*, not yet a distribution. Their total mass is

$$\sum_s \eta(s) = \sum_{t=0}^{\infty} \gamma^t \underbrace{\sum_s \mathbb{P}(S_t{=}s)}_{=1} = \sum_{t=0}^{\infty} \gamma^t = \frac{1}{1-\gamma},$$

because at each fixed time the chain is somewhere, so the time-t probabilities across states sum to one, leaving the geometric series in γ. Dividing by this total mass turns the measure into a genuine probability distribution, the *discounted visitation distribution*

$$d^{\pi_\theta,\rho_0}(s) := (1-\gamma)\,\eta(s), \qquad d^{\pi_\theta,\rho_0} \in \mathbb{R}^{|\mathcal{S}|}, \qquad \sum_s d^{\pi_\theta,\rho_0}(s) = 1.$$

So η is the unnormalized discounted visitation and d^{π_θ,ρ_0} is its normalized version; they differ only by the constant $\frac{1}{1-\gamma}$, which we carry out front. ($\eta(s)$ is read in full, as an expected discounted visit count, in Appendix B.)

5 The Theorem

Substituting $\eta(s) = \frac{1}{1-\gamma} d^{\pi_\theta,\rho_0}(s)$ and expanding ϕ_θ (every term below a d'-vector):

$$\boxed{\nabla_\theta J(\theta) = \frac{1}{1-\gamma} \sum_s d^{\pi_\theta,\rho_0}(s) \sum_a \nabla_\theta \pi_\theta(a \mid s)\, Q^{\pi_\theta}(s,a) \qquad (\in \mathbb{R}^{d'}).}$$

The constant $\frac{1}{1-\gamma}$ is positive and independent of θ, so it sets only the scale, not the direction, and is absorbed into the step size.

5.1 From a Sum to an Expectation: the Likelihood-Ratio Step

The boxed form is a weighted sum over states and actions. To turn it into something we can *sample*, we rewrite it as an expectation. The tool is the likelihood-ratio (score-function) identity, which holds coordinate by coordinate and hence as a d'-vector identity

$$\nabla_\theta \pi_\theta(a \mid s) \;=\; \pi_\theta(a \mid s)\, \nabla_\theta \log \pi_\theta(a \mid s), \qquad \nabla_\theta \log \pi_\theta(a \mid s) \in \mathbb{R}^{d'},$$

which is just $\nabla_\theta \log f = \nabla_\theta f / f$ rearranged. The point of this rewrite is that it manufactures a scalar factor of $\pi_\theta(a \mid s)$ out front. Substituting it,

$$\nabla_\theta J(\theta) = \frac{1}{1-\gamma} \sum_s d^{\pi_\theta,\rho_0}(s) \sum_a \pi_\theta(a \mid s)\, \nabla_\theta \log \pi_\theta(a \mid s)\, Q^{\pi_\theta}(s,a).$$

Now read the two weighted sums as nested expectations. The outer $\sum_s d^{\pi_\theta,\rho_0}(s)$ is an average over $s \sim d^{\pi_\theta,\rho_0}$; the inner $\sum_a \pi_\theta(a \mid s)$ is an average over $a \sim \pi_\theta(\cdot \mid s)$. Since we are integrating, the state and action become random variables, written S and A, and the integrand is the d'-vector $\nabla_\theta \log \pi_\theta(A \mid S)$ scaled by the scalar $Q^{\pi_\theta}(S,A)$:

$$\boxed{\nabla_\theta J(\theta) = \frac{1}{1-\gamma}\, \mathbb{E}_{S\sim d^{\pi_\theta,\rho_0},\ A\sim\pi_\theta(\cdot\mid S)} \big[\, \nabla_\theta \log \pi_\theta(A \mid S)\, Q^{\pi_\theta}(S,A) \,\big] \qquad (\in \mathbb{R}^{d'}).}$$

Two distributions, nested: the outer $S \sim d^{\pi_\theta,\rho_0}$ is *where you are*, the discounted visitation set by ρ_0 and the dynamics $P_{\pi_\theta}(s' \mid s) = \sum_a \pi_\theta(a \mid s)\, p(s' \mid s,a)$; the inner $A \sim \pi_\theta(\cdot \mid S)$ is *what you do there*, the policy itself.

Why the expectation form is the one that matters. An expectation is the one shape we can estimate without computing anything in closed form: draw samples, evaluate the bracket, average. Here both draws are things we already do by running the policy, S comes from the states we pass through and A from the actions we take, and the bracket $\nabla_\theta \log \pi_\theta(A \mid S)\, Q^{\pi_\theta}(S,A)$ is a d'-vector built from the analytic score and a value we can estimate from returns. This is exactly why every policy gradient algorithm starts from this line: REINFORCE, actor-critic, A2C, TRPO, and PPO all sample this expectation and differ only in how they estimate $Q^{\pi_\theta}(S,A)$ and how they control the step. (Those algorithms are the subject of separate reports; here we stop once the exact gradient is in sampleable form.)

6 Reading the Gradient

The gradient is

$$\nabla_\theta J(\theta) \;=\; \frac{1}{1-\gamma} \sum_s d^{\pi_\theta,\rho_0}(s)\, \phi_\theta(s) \;=\; \sum_s \eta(s)\, \phi_\theta(s) \qquad (\in \mathbb{R}^{d'}),$$

the same object written with the normalized d^{π_θ,ρ_0} (and the constant out front) or with the raw measure $\eta(s) = \frac{1}{1-\gamma} d^{\pi_\theta,\rho_0}(s)$, which absorbs the constant. Read in the η form, a single state's contribution $\eta(s)\, \phi_\theta(s)$ is a scalar weight times a d'-vector.

$\eta(s)$: how often we are there (a scalar). The discounted number of times the chain visits s over the whole infinite run, started from ρ_0 and following π_θ. A visit early in the run counts as a full visit; a visit at time t counts as γ^t of one. So $\eta(s)$ is large for states reached often and soon, small for states reached rarely or only far in the future.

$\phi_\theta(s)$: how much adjusting our actions there helps (a d'-vector). The gain in value at s from shifting the action probabilities at s, with the rest of the future held fixed. As a vector in parameter space it points in the direction of θ that most improves the value at s, with magnitude for how much; it is near zero where we are already acting about as well as we can at s.

The product scales each state's reinforce vector by its discounted visit count and sums. A state we visit constantly but already act well in contributes little. A state where our actions are badly wrong but we almost never reach contributes little. Only states that are both visited often and still improvable push the gradient hard.

The gradient in one sentence. Nudging θ changes how we pick actions at each state; that produces a local gain in value at each state, a direction in parameter space; and the policy gradient is the sum of those per-state gain vectors, each weighted by how often (discounted) we actually visit that state.

The effective horizon. Because $\eta(s)$ weights a time-t visit by γ^t, the visitation mass concentrates on roughly the first $\frac{1}{1-\gamma}$ steps (for $\gamma = 0.9$, about 10 steps). The states carrying that mass are set by the start distribution ρ_0 and by where the dynamics P_{π_θ} take us inside that window. So the gradient cares almost entirely about the early, reachable part of the chain, and states we cannot reach within the effective horizon are weighted to near zero.

The reading above treats $\phi_\theta(s)$ as the *local* gain at s, future held fixed, yet changing θ truly moves everything downstream. Where did that downstream effect go? The next section answers exactly that.

7 Where the Visitation Shift Went

This section is about the one feature of the result that is genuinely surprising and is the entire reason policy gradient methods exist.

7.1 The Term We Were Afraid Of

Recall the two channels from Section 1. *Channel 1*, what you do, changes the action probabilities and is captured by $\phi_\theta(s)$. *Channel 2*, where you go, changes the visitation $\eta(s)$ itself, because acting differently sends the chain down different paths. To see channel 2 explicitly, write the objective with the immediate reward pulled out. With $\bar{r}(s) := \sum_a \pi_\theta(a \mid s)\, r(s,a)$ the expected one-step reward at s,

$$J(\theta) = \sum_s \eta(s)\, \bar{r}(s),$$

and a naive product rule predicts *two* d'-vector terms:

$$\nabla_\theta J \overset{?}{=} \underbrace{\sum_s \eta(s)\, \nabla_\theta \bar{r}(s)}_{\text{channel 1: what you do}} + \underbrace{\sum_s \nabla_\theta \eta(s)\, \bar{r}(s)}_{\text{channel 2: where you go}}, \qquad \nabla_\theta \bar{r}(s),\ \nabla_\theta \eta(s) \in \mathbb{R}^{d'}.$$

The second term, $\nabla_\theta \eta(s)$, is the nightmare. Note first that it is not a $\nabla_\theta p$ term: the environment $p(s' \mid s, a)$ has no θ, so there is nothing in the environment to differentiate. The trouble is subtler. Computing $\nabla_\theta \eta$ directly means tracking how the policy-induced transition matrix $P_{\pi_\theta}(s' \mid s) = \sum_a \pi_\theta(a \mid s)\, p(s' \mid s, a)$ moves with θ and how that change propagates through every power $P_{\pi_\theta}^t$ that builds up η. Doing that needs the model p itself, even though it never needs $\nabla_\theta p$. So the naive route asks us to know the transition model and push a derivative through all of its future powers. The policy gradient theorem avoids that occupancy-derivative calculation entirely.

7.2 What the Derivation Actually Produced

The recursion did not give the two-term naive split. It gave the single form $\nabla_\theta J = \sum_s \eta(s)\, \phi_\theta(s)$, in which the scalar $\eta(s)$ appears only as an *undifferentiated* coefficient on the d'-vector $\phi_\theta(s)$. The reason is the trail it came down: in the recursion $\nabla_\theta V = \phi_\theta + \gamma P_{\pi_\theta} \nabla_\theta V$, the transition matrix P_{π_θ} entered undifferentiated, and on unrolling those undifferentiated P_{π_θ}'s became the powers $P_{\pi_\theta}^t$, then the probabilities $\mathbb{P}(S_t{=}s)$, then $\eta(s)$. At no point was a transition differentiated. So η was built from the dynamics, but built as a weight, never as a derivative of a weight.

7.3 Channel 2 Did Not Vanish: It Became the Upgrade from r to Q

It would be wrong to say the visitation shift is zero. It is real and nonzero. The honest statement is that it is not a *separate* term: it is exactly the difference between the two forms of the reinforce, the one carrying the immediate reward r and the one carrying the full action-value Q. The naive channel-1 piece carries $\bar{r}$; the derived form carries Q^{π_θ}; their difference is the entire channel-2 contribution (an identity between d'-vectors):

$$\underbrace{\sum_s \nabla_\theta \eta(s)\, \bar{r}(s)}_{\text{channel 2: } \nabla_\theta \eta} \;=\; \sum_s \eta(s) \sum_a \nabla_\theta \pi_\theta(a \mid s) \left[Q^{\pi_\theta}(s,a) - r(s,a) \right].$$

The visitation shift equals the *future-reward* part of the reinforce, $Q^{\pi_\theta} - r$, weighted by the visitation. It did not disappear; it was absorbed into the promotion of the immediate reward r to the full action-value Q^{π_θ} inside ϕ_θ. Read this identity as a *consequence of the Bellman unrolling*, not as something obtained by differentiating $\sum_s \eta(s)\bar{r}(s)$ termwise. The left side is the awkward occupancy derivative; the right side is the clean form the recursion produced. They are equal only because the unrolling is what manufactured the full Q^{π_θ} in ϕ_θ in the first place. The content of the identity is that the occupancy-shift contribution can be rewritten as future-return credit $Q^{\pi_\theta} - r$ attached to local policy derivatives.

This is why the recursive derivation never produces a $\nabla_\theta \eta$ term: it carries the full Q^{π_θ} from the very first line, and Q^{π_θ} already contains the entire discounted future. Writing Q^{π_θ} instead of r *is* accounting for channel 2. The two viewpoints, "naive split with r plus a $\nabla_\theta \eta$ correction" and "single reinforce with Q and no correction," are the same gradient.

7.4 Why This Is Legitimate, and What It Buys

The intuition behind that identity is the unrolling. "How does changing θ move my future value" dissolved into "ϕ_θ at every future state, discounted by how likely I am to reach it." Those future states are exactly what channel 2 was asking about. So the would-be $\nabla_\theta \eta$ correction, the change in *where* you go, is paid for by the fact that *when* you go there, you do the local reinforce ϕ_θ again, and that term is already in the sum, weighted by η.

The aha. You never *predict* how your policy reshapes your future state distribution; you simply *live* it. Run the new policy and the environment places you wherever it places you, and the states you find yourself in *are* the reshaped distribution. The one piece you could not have computed without a model, "how does my visitation shift," you never compute as a separate quantity, because it is already encoded in crediting each action with the full return Q^{π_θ} rather than the immediate reward, and the return is something you can sample. The recursion proved this is legitimate: letting the world move your states, while you only adjust your action probabilities at the states you land in and credit them with the full future value, gives the *exact* gradient, not an approximation.

Because no standalone $\nabla_\theta \eta$ and no $\nabla_\theta p$ survive, every ingredient of the gradient is either analytic or sampleable: $\nabla_\theta \log \pi_\theta(A \mid S) \in \mathbb{R}^{d'}$ is analytic (we built π_θ); $Q^{\pi_\theta}(S, A)$ is estimated from sampled returns or a learned critic, and is also where channel 2 now lives; and d^{π_θ,ρ_0} is never computed, since running the policy already draws states from it, with the γ^t weighting supplied by discounting the visit at time t. The visitation shift that would have demanded a model of the environment has been folded entirely into "use Q, not r," which is sampled, not modeled. That folding is what turns an apparently model-requiring gradient into one we can estimate from pure interaction, and it is why this theorem, not merely the definition of J, is what every policy gradient algorithm is built on.

A The Matrix Expansion: $\rho_0^\top P_{\pi_\theta}^t$ as a Row of State Probabilities

This appendix supports the body's claim $\rho_0^\top P_{\pi_\theta}^t = \big[\mathbb{P}(S_t{=}s_1)\ \cdots\ \mathbb{P}(S_t{=}s_n)\big]$. It is not part of the proof's logic, it is the bookkeeping that makes the step intuitive: why multiplying a power of the transition matrix by the start distribution gives the unconditional time-t state probabilities. Everything here is on the state index alone; the parameter dimension d' plays no role until the final dotting with ϕ_θ.

One step. The entry (i,j) of P_{π_θ} is the one-step probability of moving from s_i to s_j under the policy:

$$P_{\pi_\theta} = \begin{pmatrix} \mathbb{P}(S_1{=}s_1 \mid S_0{=}s_1) & \cdots & \mathbb{P}(S_1{=}s_n \mid S_0{=}s_1) \\ \mathbb{P}(S_1{=}s_1 \mid S_0{=}s_2) & \cdots & \mathbb{P}(S_1{=}s_n \mid S_0{=}s_2) \\ \vdots & \ddots & \vdots \\ \mathbb{P}(S_1{=}s_1 \mid S_0{=}s_n) & \cdots & \mathbb{P}(S_1{=}s_n \mid S_0{=}s_n) \end{pmatrix}.$$

Each row is a probability distribution over next states.

Two steps. By Chapman–Kolmogorov, $P_{\pi_\theta}^2$ holds the two-step transition probabilities:

$$P_{\pi_\theta}^2 = \begin{pmatrix} \mathbb{P}(S_2{=}s_1 \mid S_0{=}s_1) & \cdots & \mathbb{P}(S_2{=}s_n \mid S_0{=}s_1) \\ \vdots & \ddots & \vdots \\ \mathbb{P}(S_2{=}s_1 \mid S_0{=}s_n) & \cdots & \mathbb{P}(S_2{=}s_n \mid S_0{=}s_n) \end{pmatrix}.$$

t **steps.** In general $P_{\pi_\theta}^t$ holds the t-step transition probabilities:

$$P_{\pi_\theta}^t = \begin{pmatrix} \mathbb{P}(S_t{=}s_1 \mid S_0{=}s_1) & \cdots & \mathbb{P}(S_t{=}s_n \mid S_0{=}s_1) \\ \vdots & \ddots & \vdots \\ \mathbb{P}(S_t{=}s_1 \mid S_0{=}s_n) & \cdots & \mathbb{P}(S_t{=}s_n \mid S_0{=}s_n) \end{pmatrix}.$$

Left-multiply by the start distribution. The start distribution is the row vector $\rho_0^\top = \big[\mathbb{P}(S_0{=}s_1)\ \cdots\ \mathbb{P}(S_0{=}s_n)\big]$. Form the product:

$$\rho_0^\top P_{\pi_\theta}^t = \big[\,\mathbb{P}(S_0{=}s_1) \ \ \cdots \ \ \mathbb{P}(S_0{=}s_n)\,\big] \begin{pmatrix} \mathbb{P}(S_t{=}s_1 \mid S_0{=}s_1) & \cdots & \mathbb{P}(S_t{=}s_n \mid S_0{=}s_1) \\ \vdots & \ddots & \vdots \\ \mathbb{P}(S_t{=}s_1 \mid S_0{=}s_n) & \cdots & \mathbb{P}(S_t{=}s_n \mid S_0{=}s_n) \end{pmatrix}.$$

A row vector times a matrix is a row vector: its j-th entry sums the start probability of each state s_i times the probability of being at s_j after t steps from s_i,

$$\big(\rho_0^\top P_{\pi_\theta}^t\big)_j = \sum_i \mathbb{P}(S_0{=}s_i)\,\mathbb{P}(S_t{=}s_j \mid S_0{=}s_i) = \mathbb{P}(S_t = s_j),$$

which is just the law of total probability. Assembling all n entries, the product *is a single row vector of unconditional state probabilities at time* t:

$$\boxed{\rho_0^\top P_{\pi_\theta}^t = \big[\,\mathbb{P}(S_t{=}s_1)\ \ \mathbb{P}(S_t{=}s_2)\ \ \cdots\ \ \mathbb{P}(S_t{=}s_n)\,\big].}$$

This collapsed row is the whole point. We began with a matrix of *conditional* t-step probabilities (where you go given where you started) and a row of start probabilities (where you started). Multiplying them marginalizes out the start: the result no longer conditions on S_0, it is the plain distribution of the chain's position at time t. Combining it with ϕ_θ (whose row s is the d'-vector $\phi_\theta(s)^\top$) then gives the $\mathbb{P}(S_t{=}s)$-weighted sum of per-state reinforce vectors used in the body.

B The Discounted Visitation Measure $\eta(s)$

The coefficient $\eta(s) = \sum_{t=0}^{\infty} \gamma^t \mathbb{P}(S_t{=}s)$ is the single scalar object carrying the gradient's dependence on the start distribution and the dynamics. This appendix builds it from scratch and reads off its meaning. (η is purely a function of the state index; it has no parameter dimension, which is exactly why it appears as an undifferentiated weight on each $\phi_\theta(s)$.)

From probabilities to an indicator count. Fix a state s. At each time t, define the indicator random variable

$$\mathbb{1}\{S_t = s\} = \begin{cases} 1 & \text{if the chain is in } s \text{ at time } t, \\ 0 & \text{otherwise.} \end{cases}$$

Its expectation is exactly the occupancy probability, $\mathbb{E}[\mathbb{1}\{S_t = s\}] = \mathbb{P}(S_t = s)$. Substituting this into the definition and pulling the expectation outside the sum (linearity of expectation),

$$\eta(s) = \sum_{t=0}^{\infty} \gamma^t\, \mathbb{P}(S_t = s) = \sum_{t=0}^{\infty} \gamma^t\, \mathbb{E}\big[\mathbb{1}\{S_t = s\}\big] = \mathbb{E}\Big[\sum_{t=0}^{\infty} \gamma^t\, \mathbb{1}\{S_t = s\}\Big].$$

Reading the inside. For a single trajectory, the random quantity $\sum_{t=0}^{\infty} \mathbb{1}\{S_t = s\}$ is literally *the number of times that trajectory lands in* s, an integer visit count. The factor γ^t re-weights each visit by when it happened: a visit on step 0 counts as a full 1, a visit on step 1 counts as γ, a visit ten steps in counts as γ^{10}, and visits in the far future count for almost nothing.

$\eta(s)$ is the **expected discounted number of visits to** s: average, over trajectories started from ρ_0 and run under π_θ, of the time-discounted occupancy of s. The undiscounted version $\mathbb{E}[\sum_t \mathbb{1}\{S_t = s\}]$ would be the plain expected visit count; the γ^t makes it a *discounted* count in which the early part of the chain dominates.

Why discounting localizes it. The weights $1, \gamma, \gamma^2, \dots$ sum to $\frac{1}{1-\gamma}$ and decay to negligible after about $\frac{1}{1-\gamma}$ steps. So $\eta(s)$ effectively counts visits only within that early window. A state reached only deep in the chain contributes almost nothing, regardless of how often it is visited later. This is the precise sense in which the discounted objective "forgets" the far future: not by ignoring it in the return, but by weighting the visitation so that the early, start-anchored portion of the trajectory carries the mass.

Normalization. Summing the measure over all states,

$$\sum_s \eta(s) = \sum_{t=0}^{\infty} \gamma^t \underbrace{\sum_s \mathbb{P}(S_t = s)}_{=1} = \frac{1}{1-\gamma},$$

because at every fixed time the chain is *somewhere*, so the occupancy probabilities across states sum to one, and what remains is the geometric series in γ. Dividing by this total mass turns the measure into the probability distribution $d^{\pi_\theta,\rho_0}(s) = (1-\gamma)\,\eta(s)$ used in the theorem. The factor $\frac{1}{1-\gamma}$ pulled out front of the final gradient is exactly this total discounted mass.